Microsoft Campus Club - MNU

Innovation Branch | AI Track

AI & Data Science - Microsoft Machine Learning Engineer

Full Training Roadmap & Schedule

Built on the college's official Academic Calendar — 1st Semester 2026/2027

Total Training Hours: 180 accredited hours

Prepared by: Head of Projects — Microsoft Campus Club, AI Branch, Mansoura National University

Academic Year 2026 / 2027

Submitted for university accreditation as an official 180-hour training program

1. Program Overview

This is a complete AI & Data Science training program run by Microsoft Campus Club (AI Branch) at Mansoura National University, fully aligned with the college's official academic calendar (teaching weeks, quizzes, midterms, and finals). No new content is ever taught in a week that has an exam.

The program runs 2 sessions/week (Sunday & Thursday), each session 1 to 1.5 hours, with heavy hands-on practice inside the session itself, plus an applied Task after every week, a Mini Project after each major content block, and a full Graduation Project at the end of the year. Because the track starts on Sunday 11/10/2026, Term 1 also includes 2 extra Tuesday sessions (27/10 and 3/11) so that the full content and the 180 accredited hours are preserved.

Track Structure

- Term 1: Python + Excel + Data Analysis (Pandas, NumPy, Statistics, Matplotlib, Seaborn) - schedule mapped directly to the official 2026/2027 calendar you provided.
- Term 2: Machine Learning + intro Deep Learning — same scheduling logic; exact dates will be locked in once the official Term 2 calendar is released by the college.
- After each block (Python – Data Analysis – Machine Learning): a Mini Project applying everything covered.
- At the end of the year, after both terms: a full Graduation Project combining Python + Data Analysis + Machine Learning/Deep Learning, ending in a Demo Day in front of a judging panel.

Golden Scheduling Rule

No new teaching content is delivered in any week that has a Quiz, Midterm, Final, or Oral/Practical exam on the college calendar. The college's official "Revision & Mini-Projects" weeks are fully used as Mini Project Showcase weeks instead of going to waste.

2. Training Hours Breakdown (180 Hours)

To formally accredit the program as an 180-hour university training, hours are calculated across four pillars:

Item	Hours	Calculation	Notes
Live Training Sessions (Contact Hours)	57 hrs	38 sessions x 1.5 hrs	Term 1: 27 hrs (18 sessions) + Term 2: 30 hrs (20 sessions)
Weekly Applied Tasks	43 hrs	17 tasks x ~2.5 hrs	Guided self-paced work after every teaching week
Mini Projects (x3)	45 hrs	3 projects x 15 hrs	Python Automation Tool / Data Analysis Project / ML Prediction System
Graduation Project	35 hrs	4-week team capstone	Preparation + execution + Demo Day in front of a judging panel
Total	**180 hrs**	57 + 43 + 45 + 35	Matches the university's official training-hours accreditation requirement

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3. Term 1 Schedule — Python + Excel + Data Analysis

The schedule below is mapped directly onto the college's official 1st Semester 2026/2027 academic calendar. Track sessions begin on Sunday 11/10/2026, running 2 sessions/week (Sunday and Thursday), plus an extra Tuesday session in Weeks 2 and 3 (27/10 and 3/11).

Week	Session 1 (Sunday)	Extra Session (Tuesday)	Session 2 (Thursday)	Task after the week
Wk 1 *Sun 11/10 - Thu 15/10, 2026*	**Orientation + Environment Setup** Python, VS Code / Jupyter, Anaconda + full roadmap overview	—	**Python Basics I** Variables, data types, operators, Input/Output	Set up the environment + solve 5 basic exercises (calculator, odd/even, temperature converter)
EXAM WEEK — Quiz 1 (college): 17-23 Oct 2026 — no sessions				
Wk 2 *Sun 25/10 - Thu 29/10, 2026 (+ Tue 27/10)*	**Conditionals & Loops** if / elif / else, for / while	**Data Structures** Lists, Tuples, Dictionaries, Sets	**Functions** Parameters, return, *args/**kwargs, scope, lambda	Build a "Guess the Number" game + a student-grades manager using a dictionary + a small reusable functions library
Wk 3 *Sun 1/11 - Thu 5/11, 2026 (+ Tue 3/11)*	**OOP Basics** Classes, objects, attributes, methods, inheritance	**File Handling & Exceptions** Reading/writing Text, CSV, JSON files + exception handling (try/except)	**Modules, Packages & Git** Modules & packages, virtual environments (venv), intro to Git & GitHub	Build a simple "Library System" using OOP + a file-based To-Do List app + push your first repo to GitHub
EXAM WEEK — Midterm Exams + irregular-course completion (college): 7-20 Nov 2026 — no sessions				
Wk 4 *Sun 22/11 - Thu 26/11, 2026*	**Python Best Practices** PEP8, debugging, list/dict comprehensions	—	**Mini Project 1 Kickoff** Brief for "Python Automation Tool" + team assignment	Refactor previous weeks' code using comprehension + submit a Mini Project 1 plan/idea
Wk 5 *Sun 29/11 - Thu 3/12, 2026*	**Mini Project 1 Showcase + Excel for Data Analysts** Quick Mini Project 1 presentations (10-15 min), then: spreadsheet fundamentals — formulas & functions (SUM, AVERAGE, IF, VLOOKUP/XLOOKUP), tables, PivotTables & PivotCharts, and why we move from Excel to Python	—	**Pandas — Part 1** Creating, Reading, and Writing Data — Indexing, Selecting, and Assigning	Build a small sales/expenses tracker in Excel with a PivotTable summary + load a CSV dataset into Pandas and practice indexing/selecting/assigning
Wk 6 *Sun 6/12 - Thu 10/12, 2026*	**Pandas — Part 2** Summary Functions and Descriptive Stats — Handling Missing Data and Data Types	—	**Mathematical Computing with Python (NumPy)** Importance of NumPy, NumPy Arrays, Array Creation Techniques, Indexing and Slicing	Clean a messy real-world dataset (missing values, wrong types) + produce a short statistical summary + array-manipulation exercises + build a matrix calculator with NumPy
EXAM WEEK — Quiz 2 (college): 12-18 Dec 2026 — no sessions				

Week	Session 1 (Sunday)	Extra Session (Tuesday)	Session 2 (Thursday)	Task after the week
Wk 7 *Sun 20/12 - Thu 24/12, 2026*	**Statistical Analysis in Python** Introduction to Statistics — Types of Data — Descriptive Statistics — Correlation and Covariance	—	**Matplotlib & Seaborn Day** Matplotlib: Figure & Axes, Line/Bar/Histogram/Scatter, Box Plot, Subplots, Outlier Visualization, Customizing Plots — Seaborn: histplot, kdeplot, countplot, barplot, boxplot, violinplot, scatterplot, lineplot	Compute descriptive stats & a correlation matrix on a real dataset and interpret the results + build a mini visual dashboard of 5-6 charts
OFFICIAL COLLEGE "REVISION & MINI-PROJECTS" WEEK: 26 Dec 2026 - 1 Jan 2027				
Sun 27/12 — Work session with mentor support to finish the "E-Commerce Sales Analysis" project (Data Loading → Data Cleaning → Data Exploration → Statistical Analysis → Visualization → Insights)				
Thu 31/12 — Mini Project 2 full showcase + Term 1 wrap-up & student feedback				
Final / Oral / Practical Exams: 2 - 28 Jan 2027 — no sessions — teaching resumes with Term 2				

Total Term 1 sessions: 16 teaching sessions + 2 showcase sessions = 18 sessions (27 hrs of live training).

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4. Term 1 Mini Projects

Mini Project 1 — Python Automation Tool

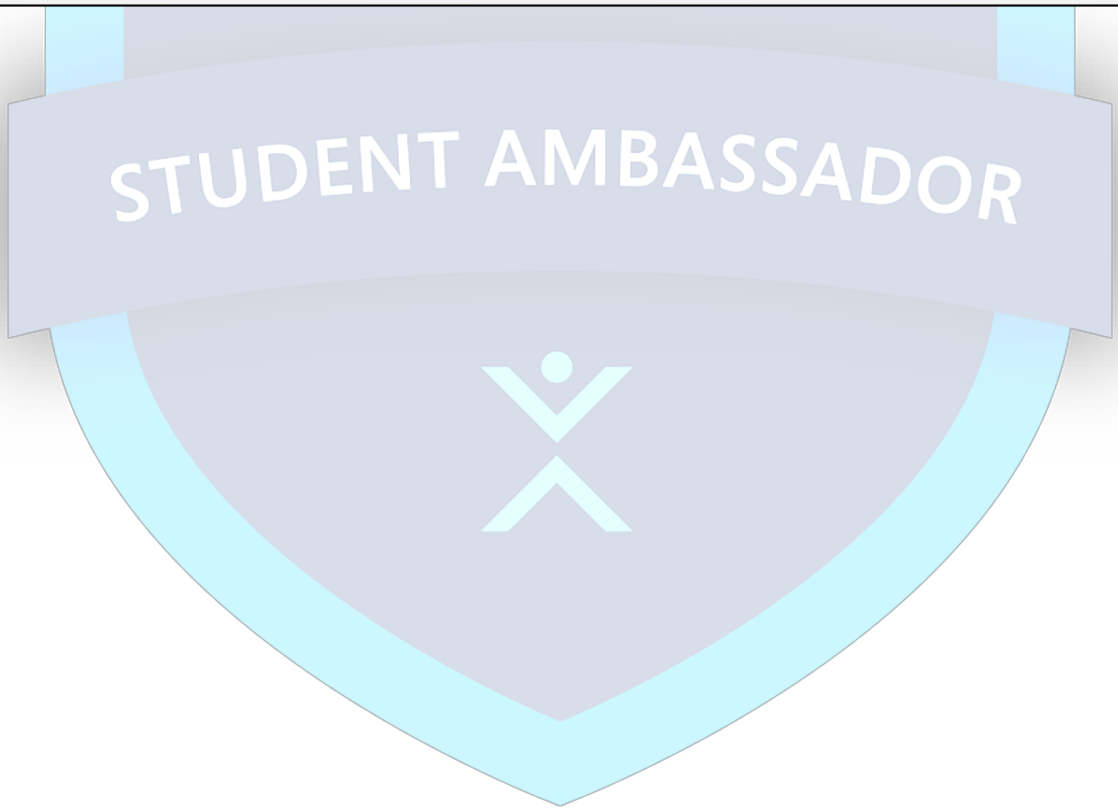
A pure-Python project (no data libraries) - e.g. a file-organizing tool, a bot that reads a file and summarizes it, or a task manager that runs entirely from the terminal.

Kickoff: end of Week 4 (26/11) - Showcase: start of Week 5 (29/11), before Data Analysis content begins.

Mini Project 2 — E-Commerce Sales Analysis (Data Analysis Project)

- Data Loading
- Data Cleaning — missing values, duplicates, data types
- Data Exploration
- Statistical Analysis — descriptive stats + correlation
- Visualization — Matplotlib & Seaborn charts
- Insights — practical findings & recommendations

Kickoff: Week 7 (introduced during the Data Analysis sessions) - Work & Showcase: official Mini-Projects week (27 & 31 Dec 2026).



5. Term 2 Schedule — Machine Learning + Deep Learning

Term 2 content (Machine Learning + intro Deep Learning) is kept as in the original roadmap, with a few suggested enhancements (see Section 6). Exact dates will be locked in once the college releases the official Term 2 2026/2027 calendar; the timeline structure (9 teaching weeks + 1 showcase week) follows the same "no teaching in an exam week" rule.

Week	Session 1 (Sunday)	Session 2 (Thursday)	Task after the week
Wk 1 *Provisional — pending official Term 2 calendar*	**What is Machine Learning?** Types (supervised / unsupervised / reinforcement) + real-world applications	**Scikit-learn Workflow** Intro to Scikit-learn + workflow (data → model → evaluation) + train/test split	Research and briefly present one real ML use-case + build your first sklearn pipeline
Wk 2 —	**Linear Regression** Theory, cost function, gradient descent	**Polynomial & Logistic Regression** Polynomial Regression and Logistic Regression	Apply Linear Regression on a housing dataset + build a Logistic Regression classifier
Wk 3 —	**KNN & Naive Bayes** K-Nearest Neighbors (KNN) and Naive Bayes	**Decision Trees & SVM** Decision Trees + intro to Support Vector Machines (SVM)	Apply KNN on a classification dataset + plot and interpret a decision tree
EXAM WEEK — Quiz 1 (college) - no sessions (pending official date)			
Wk 4 —	**Evaluation Metrics** Accuracy, precision, recall, F1-score, confusion matrix	**Cross-Validation** Cross-validation, overfitting/underfitting, bias-variance tradeoff	Re-evaluate previous models using all metrics + apply k-fold cross-validation
Wk 5 —	**Ensemble Learning** Random Forest, Bagging	**Boosting + Mini Project 3 Kickoff** AdaBoost, Gradient Boosting, intro XGBoost + Mini Project 3 kickoff	Train an ensemble model and compare it to previous models + submit a Mini Project 3 proposal
EXAM WEEKS — Midterm Exams + irregular courses (college) - no sessions (pending official date)			
Wk 6 —	**Clustering** K-Means, Hierarchical Clustering	**Dimensionality Reduction** PCA	Apply K-Means on a customer-segmentation dataset + apply PCA to reduce & visualize high-dim data
EXAM WEEK — Quiz 2 (college) - no sessions (pending official date)			
Wk 7 —	**Feature Engineering** Scaling, encoding, feature selection	**Hyperparameter Tuning** GridSearchCV, RandomizedSearchCV	Preprocess the Mini Project 3 dataset + tune the best candidate model
Wk 8 —	**Neural Network Fundamentals** Perception, activation functions, forward/backpropagation	**TensorFlow / Keras** Intro to TensorFlow/Keras + build your first Artificial Neural Network (ANN)	Implement a perceptron from scratch + train an ANN on a classification dataset
Wk 9 —	**Convolutional Neural Networks (CNN)** Intro to CNNs for image classification	**Apply CNN to the Project** Apply CNN/ML to the Mini Project 3 data + finalize the model	Build a simple CNN on MNIST / Fashion-MNIST + finalize the Mini Project 3 model
OFFICIAL COLLEGE MINI-PROJECTS WEEK (pending official date)			
Session 1 — Final work session for Mini Project 3 with mentor support			

Week	Session 1 (Sunday)	Session 2 (Thursday)	Task after the week
Session 2 — Mini Project 3 showcase — ML Prediction System (a full classification/prediction project from data cleaning to final model evaluation) + Term 2 wrap-up			
Final / Oral / Practical Exams — no sessions — Graduation Project starts right after			

Total Term 2 sessions: 18 teaching sessions + 2 showcase sessions = 20 sessions (30 hrs of live training).

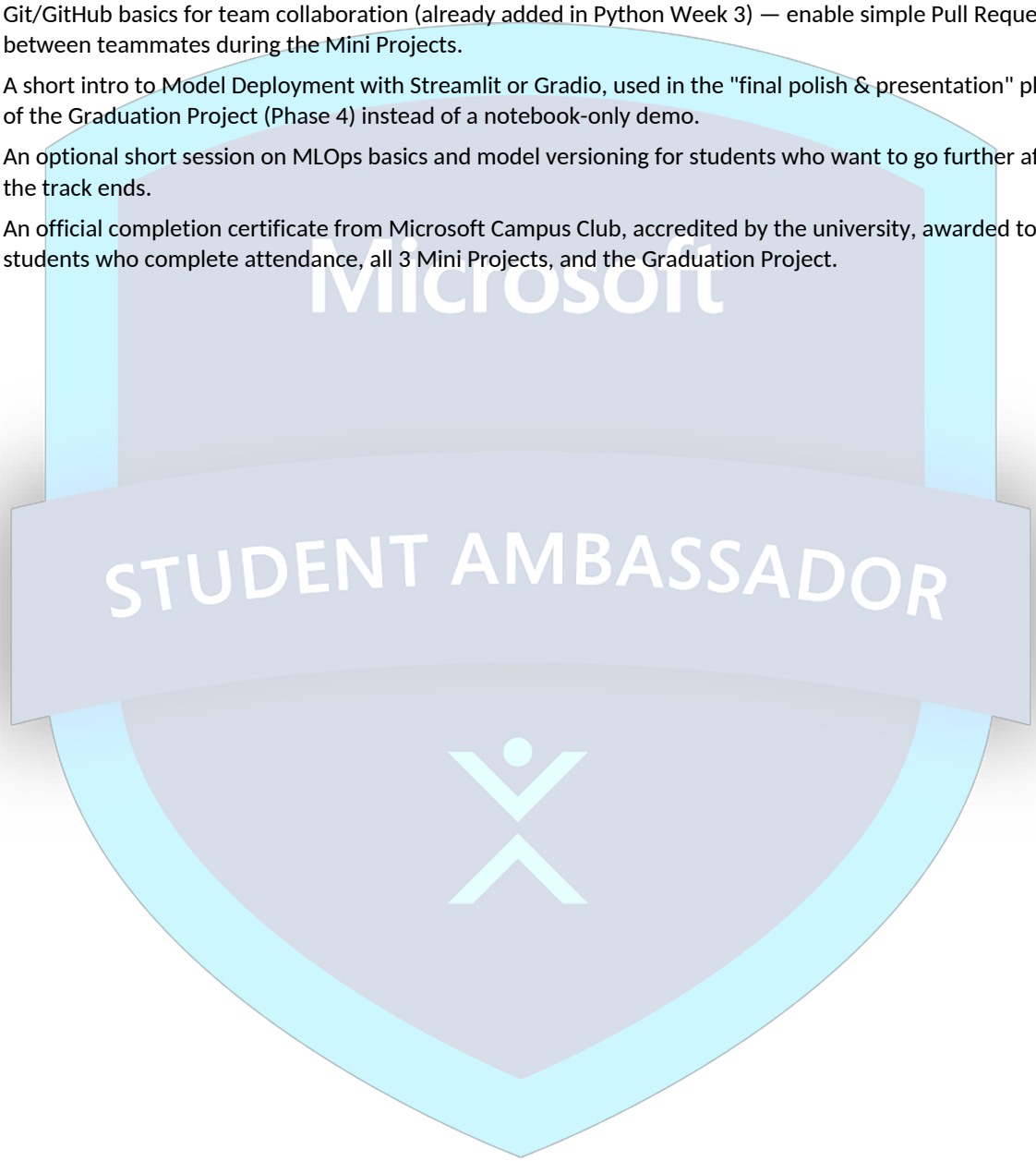
Mini Project 3 — ML Prediction System
A full classification/prediction project: choose a real dataset, clean and explore it, try more than one model (ML and/or DL), tune hyperparameters, then pick the best model with clear evaluation metrics. *Kickoff: Week 5 — Work & Showcase: official Mini-Projects week at the end of Term 2.*

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6. Suggested Enhancements (Bonus)

Since the track carries the full name "AI & Data Science - Microsoft Machine Learning Engineer", the following can be folded into existing sessions or the Graduation Project phase to add real value without disrupting the calendar-driven schedule:

- A short intro to Generative AI and Prompt Engineering, connected to Microsoft tools (Azure AI / Copilot), added to the AI-overview session at the start of Term 2.
- Git/GitHub basics for team collaboration (already added in Python Week 3) — enable simple Pull Requests between teammates during the Mini Projects.
- A short intro to Model Deployment with Streamlit or Gradio, used in the "final polish & presentation" phase of the Graduation Project (Phase 4) instead of a notebook-only demo.
- An optional short session on MLOps basics and model versioning for students who want to go further after the track ends.
- An official completion certificate from Microsoft Campus Club, accredited by the university, awarded to students who complete attendance, all 3 Mini Projects, and the Graduation Project.



7. Graduation Project — Full Capstone

After Term 2 final exams end (and before the Summer term begins), a full team capstone project kicks off where students apply everything learned during the year (Python + Data Analysis + Machine Learning / Deep Learning) to a real problem or dataset, start to finish, ending in a Demo Day where projects are presented to a judging panel.

Phase	Title	Requirements
1	Team formation & problem selection	Form teams of 3-4 + pick a real dataset/problem + write a short proposal (1 week)
2	Data preparation & analysis	Full data cleaning + EDA + initial visualization of insights (1 week)
3	Model building & evaluation	Try more than one model (ML and/or DL) + tuning + pick the best model with proper metrics (1 week)
4	Final polish & presentation prep	Build a simple demo (organized notebook or small Streamlit app) + rehearse the final presentation (1 week)
Demo Day	**Closing day**	Each team presents their project to a college/club judging panel + evaluation and prizes

Suggested Evaluation Criteria:

Criterion	Description
Data analysis quality (30%)	Depth of EDA, cleaning, and real insights extracted
Model quality (30%)	Appropriate model choice, performance, and comparison across models
Code & organization (20%)	Code clarity, repo structure, a clear README
Final presentation (20%)	Clarity of explanation, ability to answer panel questions, demo quality

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8. Program Summary & Accreditation

- 38 training sessions across both terms (18 + 20) = 57 hours of live training.
- 3 Mini Projects (Python Automation Tool, E-Commerce Sales Analysis, ML Prediction System).
- One full team Graduation Project ending in a Demo Day.
- Total of 180 accredited training hours (live training + applied tasks + mini projects + graduation project).
- Term 1 starts on Sunday 11/10/2026 and is fully locked to the official 2026/2027 academic calendar; two extra Tuesday sessions (27/10 and 3/11) keep all 18 Term 1 sessions. Term 2 dates will be locked in once the official calendar is released, keeping the same structure (no teaching in exam weeks).

